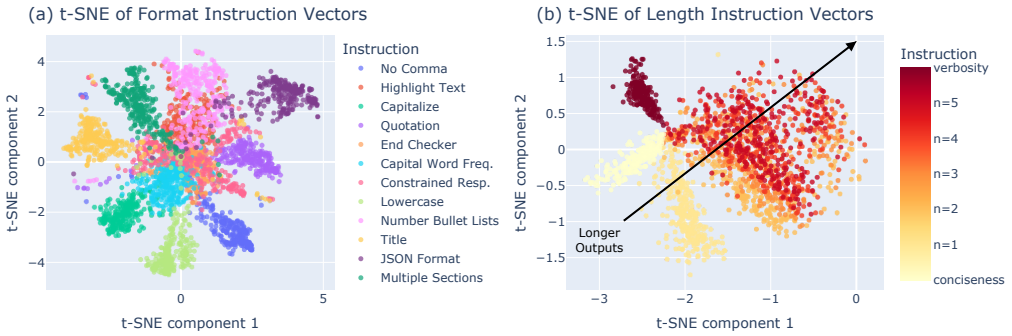

Figure 14: **t-SNE visualization of per-example instruction vectors.** (a) Format instruction vectors at layer 20 of Phi-3 for format instructions. Distinct clusters emerge for certain instructions (e.g., "No Comma," "Lowercase," "JSON Format"), while others (e.g., "End Checker," "Constrained Response") show less clear separation. (b) Vectors derived from length instructions at layer 12 of Phi-3. Vectors corresponding to concise outputs cluster on one end, while those for larger sentence counts ($n > 2$) cluster on the opposite end.

Figure 15: **Accuracy Breakdown by Format Instruction Across Models and Settings.** (a) Phi-3, (b) Gemma 2 2B IT, (c) Mistral 7B IT, and (d) Gemma 2 9B IT. Bars represent accuracy for format instructions under four settings: no explicit instruction (blue), steering without explicit instruction (red), explicit instruction without steering (green), and explicit instruction with steering (purple).

structions clustering at one end of the plot and those for concise outputs clustering at the opposite end.

I Additional Results: Format Instructions

Validation Results. Figure 16 provides examples of the results from the layer selection procedure. As detailed in Appendix E, we evaluate instruction-following accuracy and the fraction of low-perplexity outputs after steering at different layers. Panels (a) and (b) show examples obtained with Phi-3 in the no-instruction setting. In the first case ("Multiple Sections" instruction), the perplexity

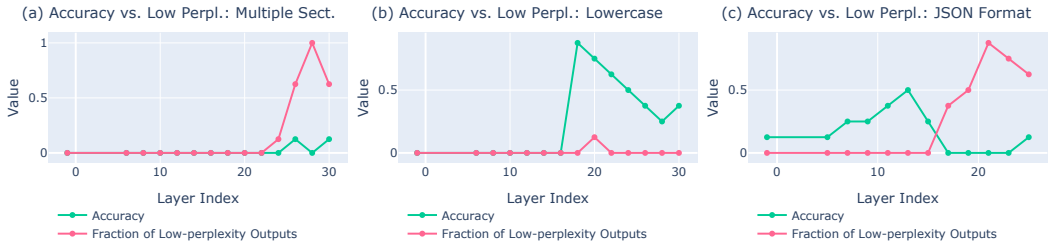

Figure 16: **Results from the layer selection procedure.** (a, b) Examples from Phi-3 in the no-instruction setting: (a) for "Multiple Sections," the perplexity check prevents selecting a layer with degraded output quality; (b) for "Lowercase," accuracy improves without quality loss. (c) For Gemma 2 2B IT with explicit "JSON Format" instructions, later layers lead to over-steering, with a degradation in both accuracy and quality.

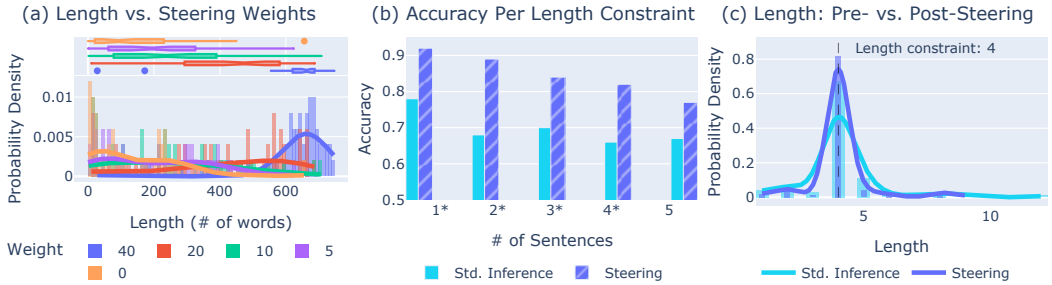

Figure 17: **Additional Results for Length Instructions.** (a) The distribution of output lengths when steering Phi-3 using a vector for verbosity, with varying steering weights. (b) Steering significantly improves accuracy for sentence-specific length constraints ($n \in 1, \ldots, 5$) in 4 of 5 cases. (c) For $n = 4$, steering shifts the output length distribution closer to the target.

check prevents the selection of a layer that would otherwise lead to a decrease in output quality, as indicated by a sharp rise in the fraction of low-perplexity outputs. Panel (c), which corresponds to the "JSON Format" instruction applied to Gemma 2 2B IT with explicit input instructions, illustrates a scenario where accuracy and output quality are correlated: steering at later layers degrades both quality and accuracy.

Accuracy Breakdown by Instruction. Figure 15 provides a breakdown of instruction-following accuracy across specific format instructions for all models and evaluation settings: no explicit instruction, steering without explicit instruction, explicit instruction without steering, and explicit instruction with steering. Across all models, steering improves accuracy for many instructions in the absence of explicit text instructions (blue vs. red bars). Notably, casing-related instructions ("Lowercase" and "English Capital") benefit strongly from steering, with Phi-3 and Gemma 2 2B IT achieving higher accuracy with steering alone than with explicit instructions (red vs. green bars). Certain instructions like "Title" and "Constrained Response" show near-perfect adherence with explicit instructions alone, leaving little room for improvement through steering.

J Additional Results: Length Instructions

Steering for Longer Outputs. In Figure 17a, we show the distribution of output lengths when the model is steered using a verbosity vector. This vector is computed from inputs containing instructions to generate verbose responses (e.g., "Provide a long answer"). Similar to our procedure for the conciseness vector in §4, we apply different steering weights to modulate verbosity. Using the same 50 base prompts as in Figure 5a, the results show that higher steering weights result in longer outputs, demonstrating that activation steering can effectively control output length by adjusting a single scalar value.

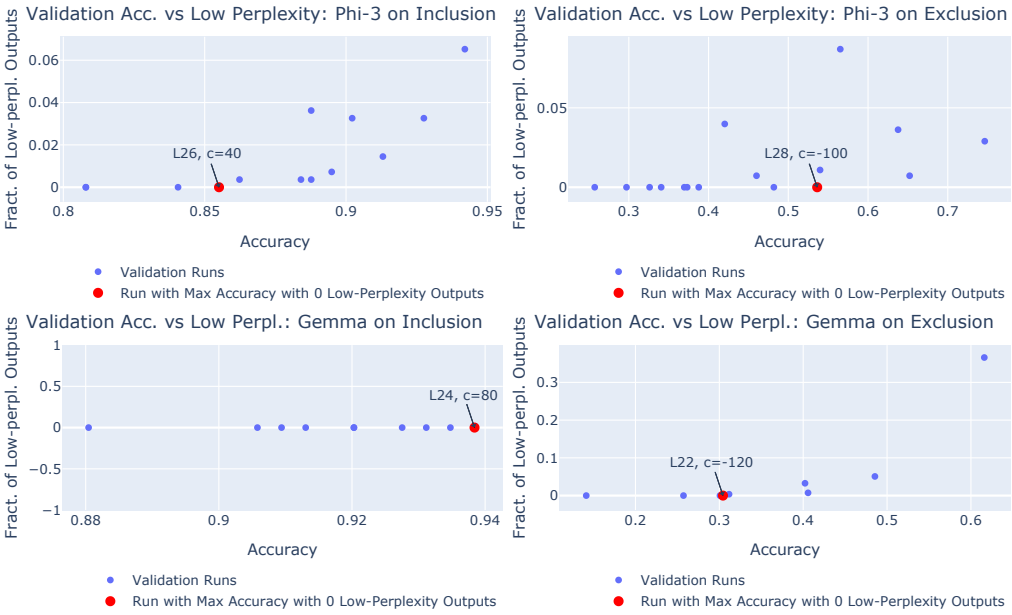

Figure 18: **Steering Layer and Weight Selection for Word-specific Instructions.** Each panel shows instruction-following accuracy (x-axis) vs. fraction of low-perplexity outputs (y-axis). Blue dots represent validation runs; red markers indicate the selected configuration with maximum accuracy and zero low-perplexity outputs. Results are shown for Phi-3 and Gemma 2 2B IT.

Table 13: **Examples of Tokens Promoted by Word-exclusion Vectors.** Projecting word-exclusion vectors onto the vocabulary results in cases where the excluded word is promoted. This suggests that word-exclusion vectors may not effectively steer the model toward avoiding these words. The vectors are computed at layer 30 of Phi-3.

Word	Top Tokens
books	␣books, ␣Books, books, ␣book, engl
congress	␣Congress, ␣parlament, ␣Legisl
Europe	␣European, ␣européen, Europe, ␣Europe
urbanization	␣urban, urban, ␣Urban, ␣urb, ␣rural
chromosomes	chrom, ␣chrom, engl, ␣genom, ␣Zob
water vapor	␣water, ␣Water, water, ␣agua
fish	␣fish, ␣Fish, fish, ␣aqu, ␣marine
adult	␣youth, ␣Youth, ␣Child, ␣Children
hypothesis	␣instead, engl, ␣invece, ␣Instead

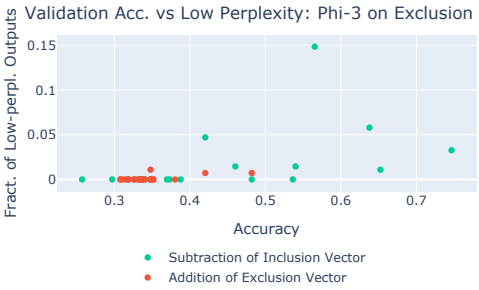

Figure 19: **Validation Accuracy vs. Perplexity Trade-off for Word Exclusion.** Subtracting word-inclusion vectors (green) outperforms adding word-exclusion vectors (red), showing the limitations of the latter.

Steering for Exact Length. We experiment with sentence-specific length instructions to evaluate the effectiveness of steering in improving adherence to explicit length constraints. Specifically, we compute steering vectors for instructions such as "Answer using n sentences" ($n \in 1, \ldots, 5$) and apply them to steer the model toward satisfying the corresponding constraints. In these experiments, we use Phi-3 and the same combination of steering layer and weight as in §4 (layer 12, $c = 20$). The results in Figure 17b show that steering significantly improves the model's adherence to the specified lengths when explicit instructions are present in the input. Notably, steering produces statistically significant improvements in 4 out of the 5 cases. Figure 17c illustrates the effect of steering on the distribution of output lengths for a specific constraint ($n = 4$ sentences). Steering shifts the distribution closer to the target length, with a sharper peak at the desired value, while still maintaining some natural variability.